

FYKE SIMON V. TONEL

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Education

STI College Caloocan	2022 – Present
Bachelor of Science in Computer Science	Caloocan City, Metro Manila
• Current Cumulative GWA: 1.27 • Best Thesis, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024-25 • Best Methods of Research Project, Bachelor of Science in Computer Science (Campus-wide – ICTE Symposium), 2024–25 • Entrant, Codefest (Local – Tagisan ng Talino), 2024–25 • President’s Honor (STI Academic Center – Academic Honors), 2022 – Present • Alumni Scholar (Awarded for High Honors in Senior High; 50% scholarship in college)	
STI College Caloocan	2020 – 2022
Senior High School (Information & Communications Technology)	Caloocan City, Metro Manila
• Graduated with High Honors • Completion of Computer Programming – Java and .NET Technology (IT in Mobile App & Web Development)	

Experience

Back-End Developer (Intern)	January 2026 – February 2026
Inspire Holdings Inc.	
• Built and deployed a NestJS payment backend on AWS (Lambda, API Gateway, RDS), integrating secure UnionBank APIs. • Reduced p95 latency from 420ms to 180ms (–57%) and CPU usage by 35% using Redis caching and token-bucket rate limiting. • Enforced API security via HMAC signing and request validation. • Delivered CI/CD pipelines and 40+ unit tests using GitHub Actions and Jest.	
Software Developer (Intern)	February 2026 – March 2026
Socia I.T. Solutions	

Projects

LootBX Mobile: Real-Time Streaming & Interactive Rewards Platform	February 2026
React Native, Expo, TypeScript, JavaScript, MongoDB	
Full-Stack Developer	
• Developed API integrations supporting real-time data flow for a mobile streaming and rewards platform. • Implemented state management logic during migration from web-based architecture to unified mobile application. • Contributed to full-stack development of a low-latency streaming system, improving responsiveness and scalability. • Applied TypeScript and modern backend patterns to improve code reliability and maintainability.	
Papa Burger Ecosystem: Multi-Platform Restaurant Operations System	February 2026 – March 2026
React TSX, PHP, PostgreSQL, Next.js, Flutter, Dart	
Full-Stack Developer	
• Built front-end architecture for a multi-module restaurant system integrating POS, driver, and admin platforms. • Developed core modules including POS system, driver portal, and franchising interface. • Designed reusable UI components and state management for scalable and maintainable frontend systems. • Integrated backend APIs to support order processing, transactions, and operational workflows.	

Projects

Flick Player: Music Player with Custom UAC 2.0	January 2026 – Present
Flutter, Dart, Rust (DSP, Audio Processing), and Kotlin	
Lead Developer – Maintainer	
• Engineered four Rust-based audio engines enabling direct USB output and bypassing Android audio pipeline constraints.	

- Architected a hybrid differential music library scanner using **Android MediaStore**, **Rust parallel traversal**, and **fingerprint caching**, reducing scan times from **~11s to 85–328ms (up to 34× faster)** across **1,000+ track SD-card libraries**.
- Optimized music library processing (scanning, metadata extraction, indexing), improving **performance on large libraries**.
- Open-source project with **600+ installs/downloads** across **GitHub and Play Store distributions**.

br41ndmg: High-Fidelity Audio Resampling Library

January 2026 – Present

Rust (DSP, Audio Processing)

Lead Developer – Maintainer

- Implemented **polyphase sinc-based resampling** for high-accuracy sample rate conversion.
- Developed configurable **FIR filters** (Hann, Hamming, Blackman, Kaiser) for precise frequency response control.
- Achieved **>100 dB stopband attenuation** and **<0.1 dB passband ripple**.
- Validated DSP performance using impulse, sine, and sweep testing with Criterion benchmarks

PASADA: Ride-Hailing & Fleet Management Ecosystem

December 2024 – November 2025

Flutter, TypeScript (Bun/Deno), React, Supabase, and Google Cloud Platforms

Lead Full-Stack Developer

- Architected a multi-platform system connecting **4 applications** (Passenger, Driver, Admin, Web) into a unified transport ecosystem.
- Developed an analytics pipeline generating **7-day predictive forecasts** for booking demand and route traffic.
- Integrated Gemini API to transform **real-time and predictive data into actionable summaries** for users and fleet managers.
- Developed a Next.js web platform with a **live ride-tracking system** using MapLibre GL and OpenStreetMap.

Mochi: Family-Shared AI Companion with Persistent Memory System

April 2026

Flutter, Node.js, SQLite, llama.cpp, Gemini API, Termux, and Kotlin

Lead Developer

- Designed a multi-user AI system with **persistent memory, mood tracking, and relationship state modeling**.
- Built a hybrid inference pipeline combining **local LLM (llama.cpp)** and **cloud fallback (Gemini API)**.
- Developed backend services for **conversation history, memory injection, and state management**.
- Deployed on a low-cost infrastructure (Termux + Cloudflare Tunnel) for **continuous availability**.

Locker: Advanced Media Hiding & Security Application

November 2025

Flutter, Dart, and Kotlin

Lead Developer – Maintainer

- Developed an Android media vault with **biometric authentication, PIN protection, and AES-256 encryption**.
- Implemented security features including **decoy vault and forced-access protection mechanisms**.
- Built a full media pipeline (import, compression, secure storage, recovery) supporting **multiple file types**.
- Released as an open-source application with **active real-world usage**.

Skills

Core Languages

Dart, Rust, Kotlin, TypeScript, Java, SQL

Systems & Backend Engineering

Node.js (NestJS, ExpressJS), Bun, Deno

PostgreSQL, SQLite, Supabase, Redis, QuestDB

WebSockets, API design, RBAC, rate limiting

Mobile & Frontend Development

Flutter, React, Next.js, Remix

Tailwind CSS, MapLibre GL JS

Low-level & Performance

Rust (systems programming, DSP, audio processing)

FFI (Flutter ↔ Rust), real-time pipelines, memory optimization

Cloud & Infrastructure

Docker, Vercel, Fly.io, Cloudflare, GCP

Event-driven systems (SQS/SNS), caching, deployment pipelines

AI & Applied ML

llama.cpp, Gemini API

Tools & Environments

Linux (Arch Linux and CachyOS), Neovim, Git, GitHub, GitLab, Postman

Android Studio, Neovim, Zed

Figma

Certifications

Learn Dart: First Step to Flutter

July 2024

Educative.io

Java Fundamentals

June 2023

Oracle